

Wf: 6.8

$$v = E(T(X)) \quad ; \quad X = X_1, \dots, X_n \text{ i.i.d.} \quad ; \quad X \sim U[3, v]$$

$$X_i \sim U[3, v] \Rightarrow \bar{X} \sim U[3, v] \Rightarrow E(\bar{X}) = \frac{3+v}{2} \Leftrightarrow v = 2E(\bar{X}) - 3$$

$$E(T(X)) = 2E(\bar{X}) - 3 = E(2\bar{X} - 3) \Rightarrow T(X) = 2\bar{X} - 3$$

$$\hat{v} = 2\bar{X} - 3 \quad \hat{v} = \frac{2}{n} \sum_{i=1}^n X_i - 3$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$